

Towards Steering without Sacrifice: Principled Training of Steering Vectors for Prompt- only Interventions

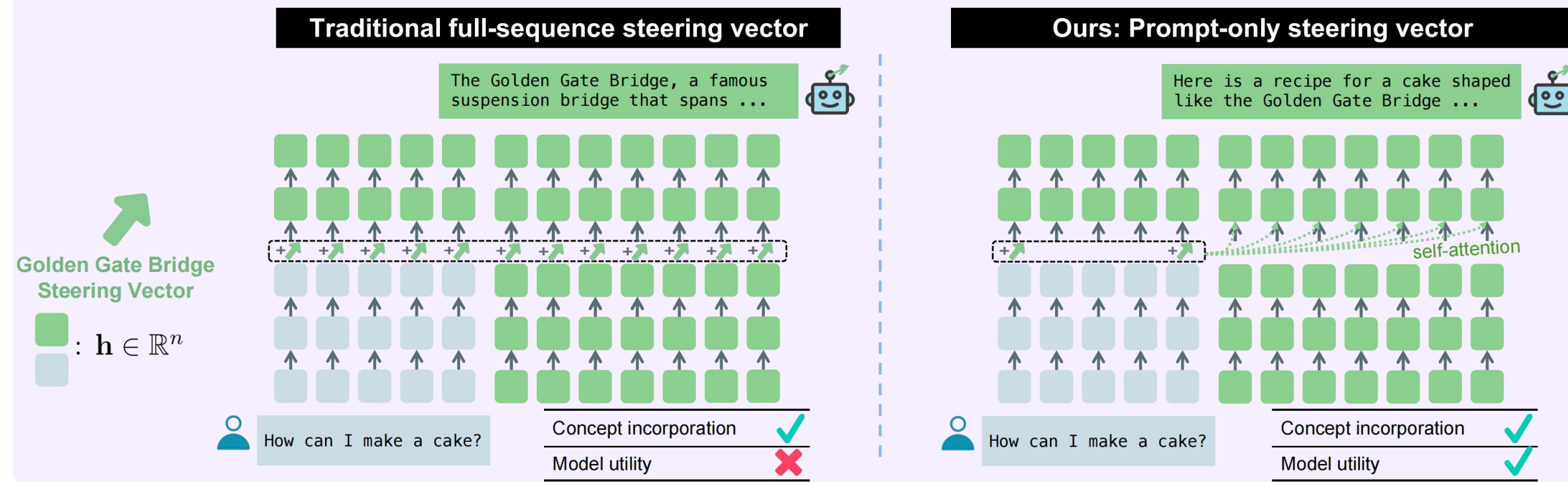
Yuntai Bao¹, Qinfeng Li¹, Xinyan Yu¹,
Ge Su¹, Wenqi Zhang¹, Liu Yan²,
Haiqin Weng², Jianwei Yin¹, Xuhong Zhang^{*1}
¹Zhejiang University ²Ant Group



ICML
International Conference
On Machine Learning

Introduction

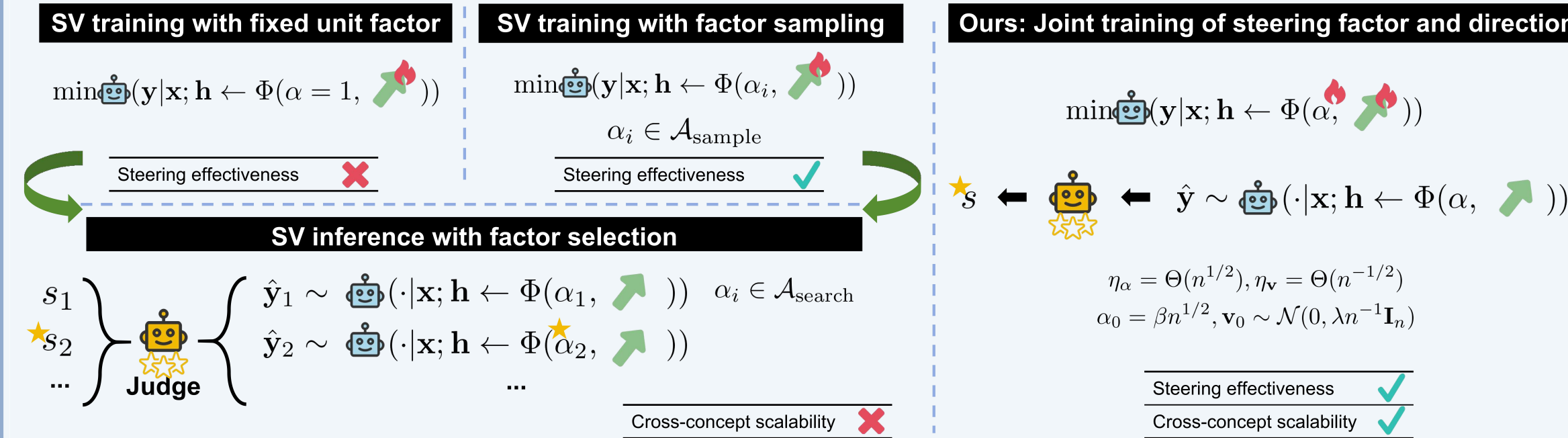
Background and Objective. *Steering vectors (SVs)* are lightweight tools for controlling model behavior, but fine-tuned SVs are still awkward to deploy: the steering factor usually has to be searched after training, and full-sequence intervention keeps perturbing the model during decoding. This paper asks **whether the factor can be learned together with the direction**, and **whether steering only (a few) prompt tokens is enough to preserve control without sacrificing generation quality**.



Contribution

- We establish a principled joint training framework for steering factors and directions based on neural network scaling theory, eliminating the need for post-hoc hyperparameter tuning.
- We introduce **Prompt-Only SV (ProSV)**, a novel intervention that restricts steering to a few prompt tokens during prefill, significantly minimizing disruption to the model's generation process.
- Experiments demonstrate that ProSV achieves superior steering-utility tradeoffs, outperforming full-sequence methods on AxBench while reducing inference overhead by 37x.

Method



Steering Vector Intervention

Add steering direction v to hidden states with factor α .

$$\tilde{h} = h + \alpha v$$

$h \in \mathbb{R}^n$: hidden states
 $v \in \mathbb{R}^n$: steering direction
 $\alpha \in \mathbb{R}$: steering factor

Prompt-only Intervention

Intervene only on prefix and/or suffix of prompts, following *representation fine-tuning*.

For a prompt with m tokens, prefix length p and suffix length s , we write “p1+s2” as shorthand for $p=1, s=2$.

Algorithm 1 Our scheme for joint training of SV factors and directions.

input Training set $\mathcal{D}$, direction learning rate $\eta_v = \Theta(-\frac{1}{2})$, factor learning rate $\eta_\alpha = \Theta(\frac{1}{2})$, direction initialization size λ , factor initialization size β , training steps T , loss function $\ell(\cdot)$

output Steering factor α_T , steering direction v_T

$v_0 \sim \mathcal{N}(0, \lambda n^{-1} I_n)$ {To ensure: $v_0 = \Theta(-\frac{1}{2})$ }

$\alpha_0 \leftarrow \beta n^{1/2}$ {To ensure: $\alpha_0 = \Theta(\frac{1}{2})$ }

$t \leftarrow 0$

while $t < T$ **do**

$(x, y) \sim \mathcal{D}$

$l_t \leftarrow \ell(p_\Phi(\cdot|x; h \leftarrow \Phi(h; \alpha_t, v_t)), y)$

$\{g_t^\alpha, g_t^v\} \leftarrow \text{Adam}(\{\nabla_\alpha l_t, \nabla_v l_t\})$

$v_{t+1} \leftarrow v_t - \eta_v g_t^v$

$\alpha_{t+1} \leftarrow \alpha_t - \eta_\alpha g_t^\alpha$

$t \leftarrow t + 1$

end while

Joint Training

Goal: To optimize steering factor α and direction v together.

The theoretical tool we use is “*network scaling theory*”, which studies the impact of hyperparameters on learning dynamics in the infinite-width limit, like μ -Parameterization.

Stability: SV-intervened feature neither explodes nor diminishes when model width goes to infinity.

$$\gamma[\alpha_t] + \gamma[v_t] = 0$$

$$\Delta \alpha_t v_{t-1} = 0,$$

$$\alpha_{t-1} \Delta v_t = 0,$$

$$(\Delta \alpha_t)(\Delta v_t) = 0,$$

$$\text{where } \Delta \alpha_t = -\eta_\alpha g_{t-1}^\alpha,$$

$$\Delta v_t = \eta_v g_{t-1}^v$$

Efficiency (feature learning): Updates to both SV factor and direction are also stable.

Experiments

Intervention Location. A small p2+s2 prefix–suffix intervention already gives the best tradeoff within our search scope, suggesting that ProSV does not need to steer the full sequence to control concepts effectively.

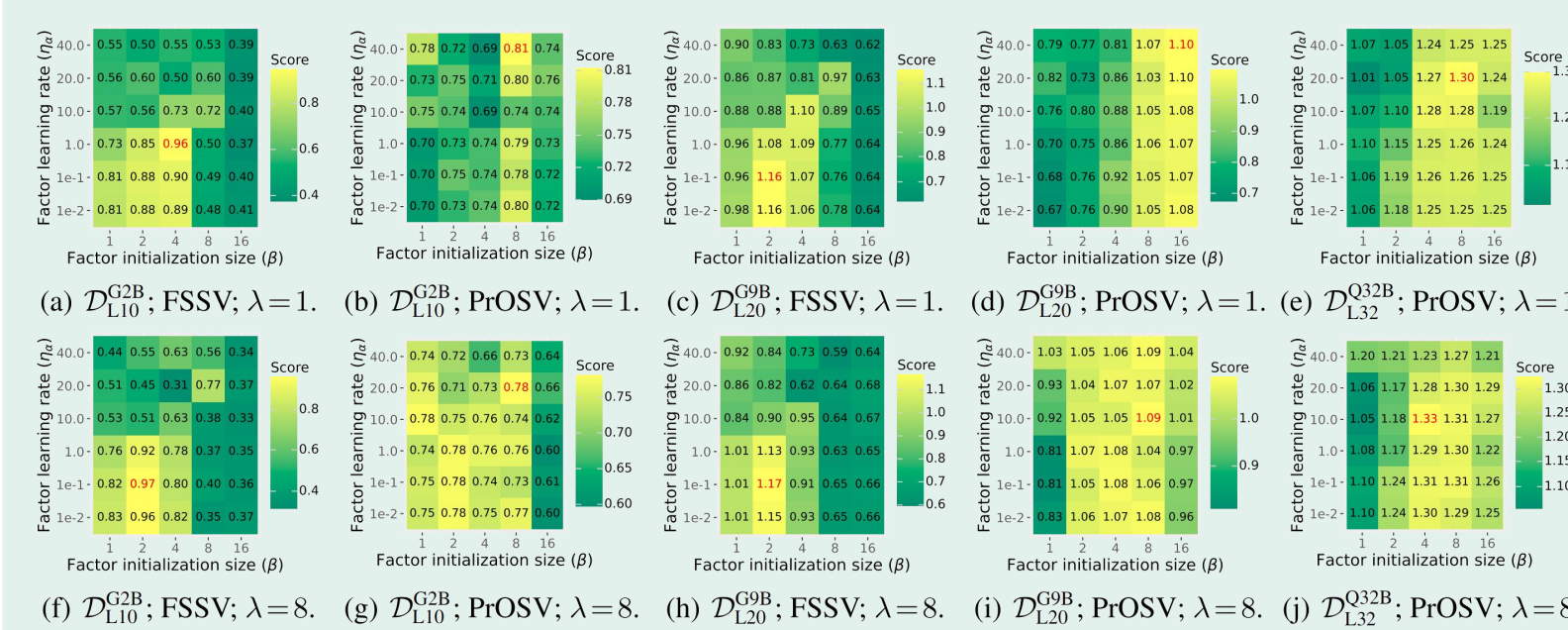
Location	G2B; L10		G9B; L20		Q32B; L32	
	O	C	O	C	O	C
FSSV	0.65	0.97	0.86	1.17	0.93	1.27
Full prompt	0.54	1.12	0.71	1.41	0.88	1.58
$ Z = 2$						
p2	0.65	0.69	0.78	0.90	0.91	1.00
s2	0.68	0.82	0.71	0.94	1.11	1.28
p1+s1	0.67	0.79	0.91	1.12	1.10	1.24
$ Z = 4$						
p4	0.59	0.63	0.73	0.85	0.94	0.97
s4	0.69	0.83	0.77	1.03	1.08	1.24
p2+s2	0.70	0.82	0.92	1.14	1.16	1.33
$ Z = 8$						
p8	0.58	0.64	0.75	0.85	0.90	0.94
s8	0.61	0.85	0.74	1.12	0.92	1.24
p4+s4	0.69	0.85	0.89	1.09	1.13	1.30

AxBench Performance. ProSV gives stronger overall steering on AxBench, with ProSV + SimPO reaching the best balance of concept control and generation quality.

Method	$\mathcal{D}_{L10}^{G2B}$	$\mathcal{D}_{L20}^{G9B}$	$\mathcal{D}_{L32}^{Q32B}$
Prompt	0.698*	1.075*	1.060
Objective: Lang.			
FSSV	0.663 [†]	0.788 [†]	0.798
+ Fixed unit factor*	0.072	0.024	—
+ Joint training	0.736	0.821	0.919
ProSV	0.758	0.859	1.049
Objective: SimPO			
FSSV (RePS)	0.756 [†]	0.892 [†]	0.947
+ Joint training	0.769	0.886	0.982
ProSV	0.803	0.905	1.102
LoReFT*	0.701	0.777	—
DiffMean*	0.297	0.322	—
SAE [‡]	—	0.546	—

Joint Training Sanity Check. We vary direction initialization size (λ), factor initialization size (β) and factor learning rate (η_α).

Best results appear with moderate factor init scale and learning rate, while init scales and learning rates that are too large/small would degrade performance.



X-axis: factor initialization size;

Y-axis: factor learning rate;

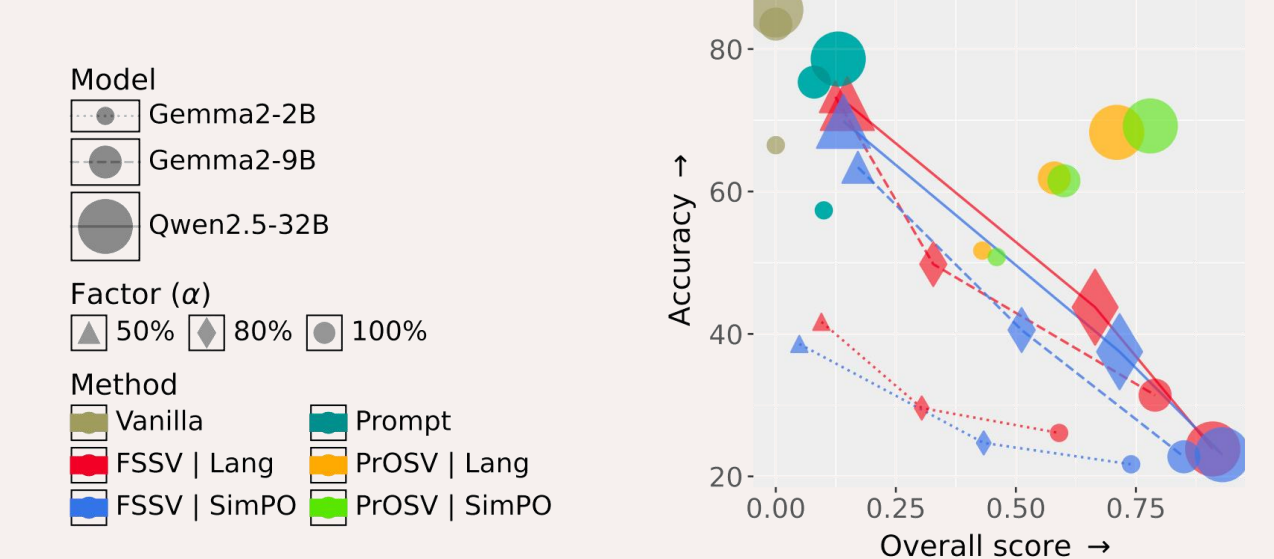
First row: unit direction initialization;

Second row: larger unit direction initialization.

Yellow means higher concept scores and green otherwise.

Analysis

Robustness–Utility Tradeoff. ProSV keeps a stronger balance under concept suppression attacks, with higher benchmark accuracy than FSSV.



Utility Preservation. ProSV retains more benchmark accuracy without weakening tinyGSM8K steering.

Method	G2B; L10			G9B; L20			Q32B; L32		
	M	G	O	M	G	O	M	G	O
M: tinyMMLU Acc.	54.0	79.0	—	74.0	93.0	—	74.0	97.0	—
G: tinyGSM8K Acc.	53.7	61.0	1.03	62.1	88.6	1.33	63.8	93.4	1.47
Objective: Lang.									
FSSV	41.5	10.7	0.66	54.2	8.6	0.64	41.1	6.6	0.75
ProSV	52.9	50.5	0.36	55.4	68.4	0.49	58.4	78.2	1.04
Objective: SimPO									
FSSV	37.8	5.6	0.70	41.3	4.2	0.75	39.2	6.9	0.90
ProSV	51.3	50.3	0.39	56.2	66.8	0.58	59.2	79.2	1.08